

Graph Generator

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Connect this into JSON files

This document shows how to use functions and apply it to data for various graphs

You can automatically extract the columns right from excel (category.year)....
losing this functionality for now

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(jsonlite)
```

```
##
## Attaching package: 'jsonlite'
##
## The following object is masked from 'package:purrr':
##
##   flatten
```

```
## Lets test with one year for now -
wincap <- read.csv("Employee_2001.csv")

## This is the problem.....

## years <- 2001:2026 ## Set years

## Employee.Info <- setNames(
  ## lapply(years, function(y) {
    ## read.csv(paste0("Employee_", y, ".csv"))
    ## }), years)

json_data <- fromJSON("Graphs.json")
```

Extract the column

```
Job.Title <- wincap$Job.Title.2001
variable_map <- list(
  Job.Title = data.frame(Job.Title = Job.Title) # ggplot needs a data frame
)
```

Graph Generator Function

```
## The column name in excel has to have this format (category.year)

plot_employee_data <- function(data, top_n = 10, x_label = NULL, title_label = NULL) {

  col_name <- colnames(data)[1] # get column name dynamically

  count_df <- data %>%
    filter(!is.na(.data[[col_name]])) %>%
    count(.data[[col_name]]) %>%
    slice_max(n, n = top_n)

  ggplot(count_df, aes(x = fct_reorder(.data[[col_name]], n, .desc = TRUE), y = n)) +
    geom_col(fill = "steelblue") +
    coord_flip() +
    labs(
      x = ifelse(is.null(x_label), col_name, x_label),
      y = "Number of Employees",
      title = ifelse(is.null(title_label), paste("Number of Employees by", col_name), title_label)
    )
}
```

```
function_map <- list(
  plot_employee_data = plot_employee_data
)

func_name <- json_data[["function"]]
arg_name <- json_data$arguments$x

func <- function_map[[func_name]]
arg <- variable_map[[arg_name]]

# Call the function
result_plot <- do.call(func, list(data = arg))
print(result_plot)
```

